THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In an AP, $S_{15} = 525$ and $S_{14} = 462$. Then $a_{15} =$ 1

(a) 987 **(b)** 67 **(c)** 59 **(d)** 63

2. The sum of the first n terms of an AP is $S_n = 5n^2 + 2n$. Its common difference is: 1

(a) 10 **(b)** 7 **(c)** 5 **(d)** 2

3. The sum of the series $14 + 19 + 24 + \cdots + 114$ is: 1

(a) 1463 **(b)** 1230 **(c)** 1344 **(d)** 2688

4. How many terms of the AP $24, 20, 16, 12, \ldots$ give sum 0? 1

(a) 12 **(b)** 13 **(c)** 26 **(d)** 14

5. If the 2nd term of an AP is 13 and the 13th term is 2, then the 15th term is: 1

(a) 15 **(b)** 26 **(c)** 0 **(d)** 1

6. The sum of the first 17 terms of an AP is 476, and its first term is 12. The 17th term is: 1

(a) 56 **(b)** 16 **(c)** 44 **(d)** 68

7. In an AP, $S_7 = 203$ and $S_6 = 147$. Then $a_7 =$ 1

(a) 350 **(b)** 47 **(c)** 56 **(d)** 65

8. The sum of the first n terms of an AP is $S_n = 1n^2 + 7n$. Its common difference is: 1

(a) 2 **(b)** 1 **(c)** 8 **(d)** 7

9. **Assertion (A):** The sequence whose nth term is $a_n = 4n - 1$ is an AP. 1

Reason (R): For this sequence $a_{n+1} - a_n = 4$ for every n.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The sum of the first 15 odd natural numbers is 225. 1

Reason (R): The sum of first n terms of an AP is $S_n = \frac{n}{2}[a + (n-1)d]$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In an AP, $a_3 + a_7 = 32$ and $a_2 + a_9 = 34$. Find a and d. 2

12. The 3rd term of an AP is 14 and the 14th term is 3. Find the 17th term. 2

13. If $S_n = 5n^2 + 3n$, find the first term and the common difference. 2

14. In an AP, $a_3 + a_7 = 50$ and $a_2 + a_9 = 56$. Find a and d. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, $S_6 = 105$ and $S_{11} = 275$. Find the AP. 3

16. Find the sum of the first 22 terms of the AP whose 5th term is 23 and 12th term is 37. 3

17. Find the sum of all integers between 0 and 120 that leave remainder 2 when divided by 8. 3

OR

In an AP, $S_5 = 75$ and $S_{10} = 325$. Find the AP. 3

18. Find the sum of the first 12 terms of the AP whose 2nd term is 13 and 5th term is 31. 3

OR

Find the sum of all integers between 50 and 500 that leave remainder 6 when divided by 13. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Kavya has a 360-page storybook. She reads 10 pages on the first day and every day she reads 2 pages more than the day before. 5

(i) How many pages does she read on the 2nd and 3rd days?

(ii) How many pages does she read on the 10th day?

(iii) How many pages does she read in the first 5 days?

(iv) In how many days does she finish the book, and how many pages does she read on the last day?

20. A loan is repaid in 30 monthly instalments; the first is ₹800 and each later instalment is ₹60 more than the previous. Find the instalment in the last month and the total amount repaid. 5

OR

A sum of ₹1410 is to be given as 12 prizes, each prize ₹15 less than the preceding one. Find the value of the first and last prize. 5

END OF QUESTION PAPER — Total: 40 Marks